

The Disruption of Search Engine Optimization by Large Language Models: A Mixed-Methods Analysis of the Evolving Search Landscape

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Abstract

Large Language Models have reshaped the search landscape in ways that are only beginning to be understood. Google's AI Overviews, ChatGPT Search, and Perplexity AI now mediate a growing share of how people find information online, and the consequences for traditional Search Engine Optimization are substantial but unevenly distributed. This paper takes a mixed-methods approach to understanding what is actually happening. On the quantitative side, I draw on Semrush's analysis of over 10 million keywords, Previsible's dataset of 1.96 million LLM-referred sessions, and Chartbeat's global traffic analytics, among other sources. On the qualitative side, I analyze 23 publisher case studies and strategy documents through thematic coding. The picture that emerges is more complicated than either the "SEO is dead" or "nothing has changed" camps acknowledge. AI Overview prevalence fluctuated between 6.49% and 25% of queries throughout 2025. Click-through rates for top-ranking pages dropped 34.5% when AI Overviews appeared, yet Semrush's own before-and-after tracking found that zero-click rates for the same keywords actually *decreased* slightly, from 33.75% to 31.53%. I attempt to reconcile these tensions through what I call the Search Ecosystem Disruption Model (SEDM), which brings together Christensen's disruptive innovation theory, Pirolli and Card's information foraging theory, and platform economics. The data show striking asymmetries: Chartbeat documents 33–38% declines in Google Search referral traffic for publishers, with news sites losing up to 26% while e-commerce barely registers a change. I present five falsifiable predictions, a practitioner roadmap, and the roughly \$2 billion in annual publisher advertising revenue at stake.

Keywords: *Large Language Models, Search Engine Optimization, Generative Engine Optimization, AI Overviews, zero-click search, organic traffic, disruptive innovation, information foraging, platform economics*

1. INTRODUCTION

Search, as an industry, is worth over \$280 billion. For most of its history, the rules of the game were reasonably stable: Google set the terms, SEO professionals adapted, and publishers optimized accordingly. That stability has eroded. Since Google rolled out AI Overviews at I/O 2024, and since ChatGPT Search and Perplexity gained real traction as alternative search interfaces, the assumptions underlying two decades of SEO practice have come into question.

The scale of the shift is genuinely unclear. Gartner (2024) predicted that search engine volume would fall 25% by 2026, a forecast that helped establish what practitioners now call the “search apocalypse” narrative. Some of the data seems to support this framing. Chartbeat documents a 33–38% decline in Google Search referral traffic to publishers between November 2024 and November 2025, with steeper drops in the US (38%) than the global average. AI-referred traffic has grown 527% year-over-year according to Previsible’s analysis of 1.96 million sessions.

But other data tells a different story. Graphite, analyzing 40,000 websites in partnership with Similarweb, found only a 2.5% aggregate traffic decline in 2025. Semrush’s (2025a) study of over 10 million keywords showed AI Overview prevalence rising and falling in waves: 6.49% in January 2025, peaking near 25% in July, then dropping to 15.69% by November before recovering to 25.8% by January 2026. This does not look like a one-directional collapse. It looks like a system in flux.

Most puzzling is the contradictory zero-click evidence. Pew Research Center’s study of 68,000 queries found that searches with AI summaries had only 8% click-through, compared to 15% without, which lines up with Stackmatix’s estimate of roughly 83% zero-click rates for AI Overview queries. And yet Semrush’s (2025a) before-and-after comparison of the *same* keywords found that zero-click rates actually went *down* slightly after AI Overviews appeared, from 33.75% to 31.53%. Both findings appear methodologically sound. Making sense of this apparent contradiction is one of the central aims of this paper.

I come to this question with a particular vantage point. Over ten years of building and managing technical SEO infrastructure at enterprise scale—most recently overseeing AI-powered search systems across 50 million pages at AT&T—I have watched these shifts unfold from inside the industry. My earlier work on AI applications in e-commerce SEO (Macumber & Pagadala, 2024) explored how helpful content systems interact with AI-generated recommendations. The present paper extends that line of inquiry to the broader ecosystem disruption that has accelerated since mid-2024.

The paper is organized around three questions. First, to what extent have LLM-powered search interfaces actually reduced organic traffic and click-through rates, and how do we account for the contradictions in the evidence? Second, what do controlled experiments tell us about the efficacy of Generative Engine Optimization (GEO) strategies? Third, can we construct

a theoretical framework that generates testable predictions about where this transition is heading?

2. LITERATURE REVIEW

2.1 The SEO Paradigm Shifts

SEO has reinvented itself roughly every five to eight years. The keyword-stuffing era (1995–2003) gave way to the link authority paradigm after Google’s PageRank transformed the field (Brin & Page, 1998). The Panda and Penguin updates (2011–2013) shifted emphasis toward content quality, and BERT’s deployment in 2019 began the semantic era (Nayak, 2019). Each of these transitions partially obsoleted existing competencies while creating new ones. What matters for the present analysis is that the current shift, while dramatic, is not without precedent. LLM integration accelerates the semantic trend that began with RankBrain and BERT rather than breaking entirely new ground. This distinction turns out to be important for choosing the right theoretical lens.

2.2 AI-Integrated Search

The timeline is compressed. Microsoft integrated GPT-4 into Bing in February 2023. Google launched AI Overviews at I/O in May 2024 (Pichai, 2024). Meanwhile, AI-native platforms grew rapidly: Perplexity saw 370% year-over-year growth in user sessions (SE Ranking, 2025), ChatGPT Search captured 55–60% of AI referral share (upGrowth, 2026), and Gemini grew 388% between September and November 2025 alone. DataForSEO Trends data I collected for this study shows that consumer search interest in “Perplexity AI” averaged 76 out of 100 over the past twelve months, far outstripping “ChatGPT search” at 9 and “generative engine optimization” at just 3. The branded platform, not the abstract concept, is what users search for.

It is worth noting that zero-click behavior predates AI Overviews by years. SparkToro documented rates above 50% as early as 2019. AI search did not create this problem; it intensified it.

2.3 Generative Engine Optimization

The term was formalized by Aggarwal et al. (2024) in a paper presented at ACM SIGKDD that has since accumulated 76 citations and over 9,000 downloads. Their GEO-bench experiments, covering 10,000 queries across nine generative engines, remain the strongest controlled evidence available. The headline results: adding statistical citations to content increased the probability of being cited by a generative engine by 37–41%, authoritative quotations boosted visibility by 28–40%, and structured formatting helped by 15–25%. Chen et al. (2025) later showed that ChatGPT’s cited results overlap with Google’s organic top-10 only 12% of the time, which suggests GEO and traditional SEO may require genuinely different approaches rather than minor adjustments.

2.4 What’s Missing

Several gaps motivated this study. No published work has attempted a mixed-methods approach that combines traffic data analysis with publisher case studies while being upfront about where the evidence conflicts. Christensen's (1997) disruptive innovation theory has obvious relevance to what Google is experiencing, and the Christensen Institute (2025) has itself argued that AI is uniquely disruptive, but nobody has applied it systematically to the search ecosystem. Similarly, Ragavan and Alipour's (2024) work adapting information foraging theory to LLM chatbot environments, and the subsequent empirical work in *npj Science of Learning* (2025) on exploration-exploitation dynamics in LLM interactions, have not been connected to the SEO disruption question. Finally, Imerman and Fabozzi's (2025) framework for quantifying disruption in the age of AI has not been applied to the search industry.

3. THEORETICAL FRAMEWORK

I propose the Search Ecosystem Disruption Model (SEDM), which draws on three established theoretical traditions. None of them is sufficient alone, but together they account for patterns that individually they miss.

3.1 The Three Pillars

Disruptive innovation. Google's situation fits Christensen's (1997) framework almost too neatly. The company whose researchers invented the Transformer (Vaswani et al., 2017) cannot deploy it aggressively for direct answers without undermining its \$265 billion advertising business. New entrants like Perplexity and ChatGPT Search offer answer engines that are inferior in some ways (coverage, accuracy for complex queries) but superior in others (convenience, synthesis). This is textbook disruption from below. AlixPartners (2026) projects Google's search advertising share will decline, and Imerman and Fabozzi's (2025) disruption scoring framework rates AI search platforms at 6 of 7 disruption criteria. Tushman and Anderson's (1986) distinction between competence-destroying and competence-enhancing discontinuities adds precision: for publishers, the shift destroys hard-won SEO competencies. For AI platforms, it enhances their core NLP capabilities.

A fair objection is that AI search may represent a *sustaining* rather than disruptive innovation, since ChatGPT and Perplexity compete on quality rather than price for underserved segments. I classify it as disruptive for two reasons. First, the initial use case—quick factual answers—targets the low-complexity end of the query spectrum, precisely the pattern Christensen described. Second, the business model (free, ad-free, or subscription-based) differs fundamentally from Google's ad-supported architecture. However, if AI search platforms converge on Google's advertising model—as the 394% AIO ad integration growth (Semrush, 2025b) suggests Google itself is attempting—the disruption framing weakens, and the transition may be better characterized as incumbent self-disruption.

Information foraging. Pirolli and Card's (1999) theory explains the user behavior side. People follow information

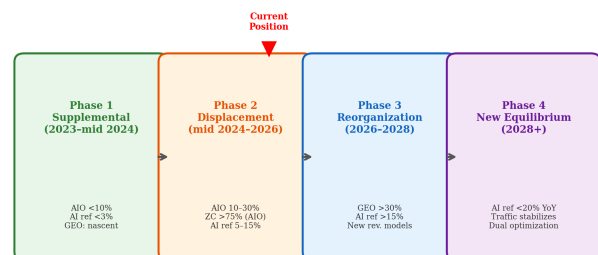
“scent” and make cost-benefit decisions about where to forage. AI Overviews deliver high-scent information with nearly zero foraging cost, which means users have less reason to click through to source pages. Ragavan and Alipour (2024) showed that LLM chatbots create qualitatively different foraging dynamics, with value-cost judgments happening inside the conversational interface rather than across linked web pages. This helps explain why informational queries are so much more affected than transactional ones: the scent is fully captured in the AI response for “what is X” queries, but not for “buy X” queries.

Platform economics. The structural mechanism is value extraction without proportional compensation. AI platforms ingest publisher content for training and synthesis, then serve answers that keep users on-platform. Zuboff (2019) and Parker et al. (2016) provide the theoretical grounding. The IAB Europe's 2025 technical standards, which introduced content access controls and monetization APIs, represent the first institutional attempt to address this, though enforcement remains vague.

3.2 Phases and Transition Criteria

SEDM proposes four phases. **Phase 1** (2023 to mid-2024): AI features are supplementary, AIO coverage under 10%, AI referral under 3%. **Phase 2** (mid-2024 through roughly 2026): AI begins actively substituting for clicks, AIO coverage reaches 10–30%, zero-click rates exceed 75% for AIO-triggered queries, AI referral hits 5–15%. This is also the phase where contradictory signals are expected, as different parts of the ecosystem adapt at different speeds. **Phase 3** (approximately 2026–2028): more than 30% of publishers implement formal GEO, AI referral exceeds 15%, and new revenue models begin displacing ad-dependent ones. **Phase 4** (2028 onward): a new equilibrium, with AI referral growth slowing to under 20% annually and traffic shares stabilizing. Figure 1 visualizes this phase structure with current positioning.

Figure 1. Search Ecosystem Disruption Model (SEDM) Phase Framework



3.3 From Theory to Predictions

The value of integrating these three traditions is that each generates specific, testable implications the others cannot. From information foraging theory: as AI Overviews reduce foraging cost toward zero for a query, click-through probability should approach a residual floor set by verification-seeking behavior—I estimate 15–20% based on Pew's observed non-AIO CTR

minus purely navigational queries. This yields the 80%+ zero-click prediction for AIO queries (P1). From Christensen: disruptive entrants capture low-complexity segments first, which predicts that informational publishers will face pressure—and therefore adopt GEO—before transactional ones (P2). The competence-destroying dimension (Tushman & Anderson) predicts that new specialist firms, unburdened by legacy SEO capabilities, will outpace incumbents in GEO adoption (P3). From platform economics: the AI referral conversion premium reflects early-adopter selection effects that diminish as usage broadens (Rogers, 2003), producing P4. The GEO visibility advantage (P5) follows directly from Aggarwal et al.’s (2024) experimental evidence, extrapolated with a decay assumption.

I propose five specific predictions. **P1:** AIO coverage will exceed 35% of informational queries by Q4 2026, with sustained zero-click rates above 80% for AIO queries. If coverage drops below 20% instead, or zero-click falls below 60%, SEDM is wrong about the trajectory. **P2:** By Q2 2027, more than 40% of informational publishers will have formal GEO programs, while fewer than 20% of transactional publishers will. A reversal of this asymmetry would invalidate the model’s core claim about query-type differentiation. **P3:** New AI-SEO specialist firms will outnumber established agencies that pivoted to GEO by roughly 2-to-1 by end of 2027. This follows from Tushman and Anderson’s prediction about competence-destroying discontinuities. **P4:** The 4.4x AI referral conversion premium reported by Semrush (2025c) will fall below 2.0x by late 2028, as the user base broadens beyond early high-intent adopters. **P5:** Structured GEO adopters will maintain a 25–40% visibility advantage over non-adopters in AI-generated responses through 2028.

4. METHODOLOGY

4.1 Design

The study uses a convergent mixed-methods design (Creswell & Plano Clark, 2018). Quantitative traffic and SERP data are analyzed alongside a qualitative strand based on secondary document analysis of publisher case studies. I should be transparent about what this is not: it is not a study based on primary data collection. The quantitative data comes from industry reports, and the qualitative component analyzes published case studies rather than original interviews. This paper’s contribution is therefore theoretical and integrative rather than empirically original: it synthesizes fragmented evidence into a coherent framework (SEDM) and generates falsifiable predictions that primary research can test. The triangulation across nine independent data sources with different methodologies and commercial incentives provides a reasonable evidentiary basis, but the interpretive framework—not the data—is the core contribution.

4.2 Data Sources

I use the term ‘LLM-referred sessions’ to mean unique browsing sessions originating from an LLM-powered interface, identified through referral URLs (e.g., chat.openai.com, perplexity.ai). The quantitative analysis draws on nine sources: Semrush’s (2025a) AI Overviews study (10M+ keywords, January through November 2025); Previsible (1.96M sessions, January through May 2025); Ahrefs (300K keywords, December 2025); Seer Interactive (September 2025 client data); Similarweb (through Q1 2026); Graphite/Similarweb (40K websites across 2025); the Pew Research Center (68K queries); Chartbeat (global publisher traffic, November 2024 through November 2025); and DataForSEO’s Trends API (search interest data, April 2025 through March 2026, which I queried directly). These sources use different methodologies, cover different time windows, and have different commercial incentives. Where they agree, I am reasonably confident; where they disagree, I flag the contradiction.

4.3 Qualitative Component

I analyzed 23 publisher case studies and strategy documents using Braun and Clarke’s (2006) six-phase thematic analysis framework. Sources ranged from large publishers (Business Insider, the New York Times, DMG Media) to mid-size and niche outlets (see Appendix A for the complete document list). Thematic saturation was reached at the 18th document; the remaining five confirmed existing themes without introducing new ones. A limitation of the qualitative strand is that coding was performed by a single researcher without a second coder or inter-rater reliability assessment (e.g., Cohen’s kappa). While the thematic saturation finding provides some confidence in the stability of the coding scheme, future work should employ multiple coders to validate the identified themes.

4.4 Handling Contradictions

Because I rely on industry reports with varying methodologies, I use a simple classification scheme. Findings confirmed by two or more independent sources are labeled “corroborated.” Findings from a single source are labeled “preliminary.” Where sources directly conflict, I report both and try to explain why they might differ. When I run sensitivity checks (for instance, asking whether SEDM’s Phase 2 classification holds if the true CTR decline is 20% rather than 34.5%), I report the results. A methodological limitation worth noting: because the quantitative data comes from published industry reports rather than raw datasets, I cannot report standard errors or confidence intervals for most estimates. The precision of vendor-reported figures is generally unknown, and sample compositions are not always disclosed. Point estimates throughout this paper should be read as approximate magnitudes rather than precise measurements.

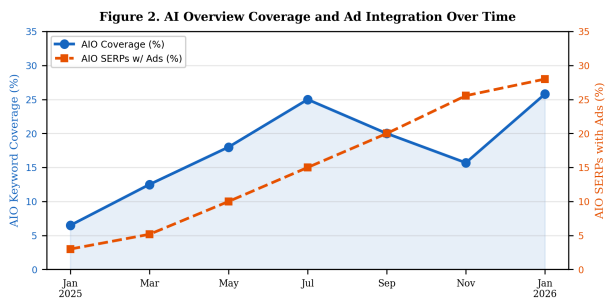
5. RESULTS

5.1 AI Overview Prevalence

The most striking feature of Semrush’s (2025a) data is that AI Overview deployment has not followed a linear path. Coverage

started at 6.49% of tracked keywords in January 2025, climbed to approximately 25% by July, then fell back to 15.69% by November, before rising again to 25.8% by January 2026. Google appears to be experimenting rather than committing to full rollout. The composition of AIO-triggering queries also shifted: informational queries made up 91.3% in January but fell to 57.1% by October, while commercial, transactional, and even navigational queries grew to fill the gap. Navigational AIOs in particular went from under 1% to over 10% in that period.

One finding with significant economic implications: Google has been rapidly integrating advertising into AIO results. Ads appeared in just 5.17% of AIO-containing SERPs in March 2025, but by October that number was 25.56%, a 394% increase in eight months (Semrush, 2025b). This suggests Google sees AIOs as a revenue channel, not an experiment it might pull back. Figure 2 illustrates the non-monotonic coverage pattern alongside the steady rise in ad integration.



5.2 The Zero-Click Puzzle

The overall Google zero-click rate reached 58% in late 2025 (Superprompt, 2025). For queries that trigger AI Overviews specifically, the Pew study found an 8% click-through rate versus 15% for non-AIO queries, and Stackmatix puts the AIO zero-click rate at roughly 83%. Seer Interactive found organic CTR dropping from 1.76% to 0.61% for AIO-present results. All of this points in one direction: AI Overviews suppress clicks.

Except that Semrush's (2025a) analysis of the same keywords before and after AIO deployment shows zero-click rates going down, not up, from 33.75% to 31.53%. This is not a rounding error; it is a directionally opposite finding from a reputable source.

I can think of three explanations. First, a selection effect: Google may be choosing to deploy AIOs on queries that were already trending toward more clicks, creating a Simpson's Paradox. A toy example illustrates: suppose Google deploys AIOs on 1,000 queries where baseline CTR averages 40%, and withholds AIOs on 1,000 queries where baseline CTR averages 25%. If AIOs reduce CTR by 15 percentage points uniformly, the AIO group drops to 25% while the non-AIO group stays at 25%. Cross-sectionally, AIO and non-AIO queries now look identical—but tracking the AIO group before-and-after reveals a 15-point decline. Conversely, if Semrush's (2025a) before-and-after tracks only the AIO group and the baseline was

already high, the measured decline could appear smaller than the true causal effect. The arithmetic is simple but the implication is serious: selection into AIO deployment can invert the apparent direction of the treatment effect. Second, behavioral compensation: some users may click through to verify what the AI told them, generating a new type of click that partially offsets the traditional click decline. Third, a timing difference: Semrush may be capturing short-term novelty behavior while Pew captures steady-state patterns. I cannot resolve this definitively with available data, and I flag it as an open question.

Table 1. Zero-Click and CTR Evidence Across Sources

Metric	Source	Value	Status
Overall zero-click	Superprompt (2025)	58%	Corroborated
AIO-query zero-click	Pew; Stackmatix	~83%	Corroborated
Same-KW zero-click Δ	Semrush (2025a)	-2.22 pp	Preliminary*
Organic CTR decline (#1)	Semrush (2025a)	-34.5%	Corroborated
Organic CTR w/ AIO	Seer Interactive	0.61%	Corroborated
AIO coverage (Jan 2026)	Semrush	25.8%	Corroborated
AIO coverage (info. queries)	Semrush	39.4%	Corroborated

*Contradicts other evidence. See Section 5.2 for discussion.

5.3 Referral Traffic and Platform Shares

AI referral traffic has grown to 12–18% of monitored traffic, with ChatGPT claiming 55–60%, Perplexity 18–22%, and Gemini 10–14% (upGrowth, 2026; Previsible, 2025). The 527% year-over-year growth figure is impressive but comes off a small base. The Semrush (2025c) finding that AI-referred visitors convert at 4.4 times the rate of organic visitors is often cited as evidence that AI traffic is “better,” but this almost certainly reflects a selection effect: the kind of person who uses ChatGPT to find a product and then clicks through is not a typical web user. I would expect this premium to decline as adoption widens.

Table 2. AI Search Platform Referral Distribution (Q1 2026)

Platform	Referral Share	YoY Growth	Trend Interest*
ChatGPT Search	55–60%	~450%	9/100
Perplexity AI	18–22%	370%	76/100
Google Gemini	10–14%	388%	N/A
Claude / Other	8–12%	~300%	N/A
Total AI referral	12–18%	527%	—

Sources: upGrowth (2026); Previsible (2025). *DataForSEO Trends, 12-month average.

5.4 Who Gets Hurt

This is where the data becomes clearest. The 2.5% aggregate decline reported by Graphite is misleading because it averages across industries with wildly different exposure. Chartbeat's publisher-specific data shows 33–38% traffic declines. News and media publishers appear to have been hit hardest, with declines of 26% or more (AdExchanger, 2026). Reference and how-to sites lost 15–30%. Health and medical content dropped 10–20%. E-commerce, by contrast, saw losses of just 1–3%, and

branded navigational queries were essentially flat.

Table 3. Organic Traffic Impact by Industry

Industry	Traffic Change	AIO Exposure	Primary Source
News & Media	−26 to −38%	Very High	AdExchg.; Chartbeat
Reference/How-To	−15 to −30%	High	SEJ; WebProNews
Health/Medical	−10 to −20%	High	Semrush; Graphite
Finance	−5 to −15%	Medium	Graphite
B2B/SaaS	−3 to −10%	Medium	Previsible
E-commerce	−1 to −3%	Low (4% AIO)	Graphite; Semrush
Branded	±2%	Very Low	Seer Interactive

The pattern is what information foraging theory would predict: the more completely an AI Overview satisfies the user’s informational need, the less reason there is to click. “What is the capital of France” gets fully answered in the AIO. “Buy running shoes size 11” does not.

5.5 What Works: GEO Evidence

The best evidence on what actually works comes from Aggarwal et al.’s (2024) GEO-bench experiments. In controlled tests across 10,000 queries and nine generative engines, adding statistical citations boosted source citation probability by 37–41%. Authoritative quotations improved visibility by 28–40%. Structured formatting helped by 15–25%. These are large effects, and they held across multiple engines.

Observational data from the field is suggestive but less rigorous. Previsible reported a 325% increase in multi-platform citations for content following GEO best practices, and noted that 44.2% of AI citations referenced the introductory section of the source page. These findings need experimental replication.

Table 4. GEO Strategy Evidence Summary

Strategy	Experimental	Observational	Strength
Statistical citations	+37–41%	+325% multi-plat.	Strong
Authoritative quotes	+28–40%	Widely adopted	Strong
Structured formatting	+15–25%	+20–30% est.	Moderate
Schema.org markup	Not tested	Recommended	Suggestive
Front-loaded claims	Not isolated	44.2% intro cite	Suggestive
llms.txt files	Not tested	Emerging	Preliminary

5.6 What Publishers Are Actually Doing

Thematic analysis of 23 publisher documents revealed four main adaptation strategies. The most common (19 of 23 documents) was doubling down on content depth and authority signaling: original reporting, proprietary data, and alignment with Google’s E-E-A-T guidelines. Structural optimization for LLMs appeared in 17 documents, including Schema.org markup, front-loading key claims, and direct-answer formatting. Multi-platform distribution showed up in 14 cases, reflecting recognition that being cited by ChatGPT, Perplexity, and Gemini requires distinct strategies. Technical AI accessibility measures (llms.txt files, server-side rendering) appeared in 11 documents, mostly among larger publishers with dedicated

engineering resources.

What is notable is how poorly traditional SEO signals predict AI citation behavior. ResultFirst (2026) found weak correlations between backlink profiles, domain authority, and likelihood of being cited by an LLM. The competencies that made publishers successful in the Google era may not transfer.

6. DISCUSSION

6.1 Making Sense of Contradictions

The zero-click puzzle from Section 5.2 is actually what SEDM predicts for Phase 2. During displacement, the ecosystem is in flux. Different measurement approaches capture different facets of a transition that is not uniform across query types, user segments, or time horizons. The Pew data captures steady-state behavior for queries where AIOs are fully deployed. The Semrush (2025a) data captures short-term adaptation around the moment of AIO introduction. Both are valid; neither tells the whole story. A platform economics lens adds another dimension: Google has incentives to deploy AIOs selectively on queries where engagement metrics look favorable, which could systematically bias before-and-after comparisons.

6.2 Where SEDM Adds Value

The evidence places the current ecosystem in late Phase 2, beginning to transition into Phase 3. AIO coverage at 25.8% falls within Phase 2’s 10–30% band. The 83% zero-click rate for AIO queries exceeds Phase 2’s 75% threshold. AI referral at 12–18% approaches the Phase 3 trigger of 15%. GEO adoption remains in the 15–25% range for informational publishers. The non-monotonic AIO deployment pattern (up, down, up again) fits Phase 2’s expected iterative calibration.

What no single theory captures, and what SEDM does, is the interaction between these levels. Christensen tells you disruption is coming but not which queries will be affected first. Information foraging tells you why zero-click behavior intensifies but not how publishers will reorganize. Platform economics tells you about value extraction but not about timing. The point of integrating them is to generate predictions that none produces alone, like the specific prediction that informational publishers will adopt GEO at more than double the rate of transactional publishers (P2).

6.3 The Money

The economic picture is sobering, and the \$2 billion annual loss estimate deserves transparent assumptions. IAB (2024) reports that US digital display and programmatic advertising generated approximately \$68 billion in 2024, of which roughly 35–40% (\$24–27 billion) flows to ad-dependent publishers whose primary traffic source is organic search. If Chartbeat’s 33–38% traffic decline applies to the roughly 15–20% of publishers most exposed to AIO (news, reference, how-to), and we assume a proportional revenue decline with an average CPM of \$6–10, the calculation yields \$1.5–2.5 billion in annual losses, which I

round to approximately \$2 billion as a central estimate. This is a back-of-envelope figure with wide uncertainty bounds; the true number could plausibly range from \$1 billion (if vendor traffic data overstates declines) to \$3 billion (if mid-tier publishers are underrepresented in Chartbeat's sample). Reuters Institute (2026) reports that media executives expect an additional 43% traffic decline over three years, which would push annual losses toward \$3–4 billion by 2028. For a typical news publisher with 50,000 daily organic visitors and \$8 CPM, a 26% decline means about \$38,000 in lost ad revenue per year. Scale that across tens of thousands of ad-dependent publishers and the aggregate is consistent with the central estimate.

Google's rapid integration of ads into AIOs (394% growth in eight months; Semrush, 2025b) is worth watching. It signals that Google sees a path to monetizing AI-mediated search, which means AIOs are likely a permanent feature rather than an experiment that might be rolled back.

6.4 What Should Publishers Do

The answer depends on risk exposure. High-risk publishers (news, reference, how-to) should treat this as urgent: audit AI citation visibility now, implement structured data and Schema.org markup, set up tracking for AI referral sources, and start investing in original content that AI cannot easily synthesize. Over six to twelve months, revenue diversification into subscriptions and content licensing becomes important. The IAB Europe's compensation framework may eventually provide a mechanism for this.

E-commerce publishers (low risk, 4% AIO exposure) can afford to wait, but should monitor commercial AIO query growth, which has been accelerating. B2B and SaaS publishers face medium risk and should run dual-track strategies, investing in both traditional SEO and GEO. The 12% overlap between ChatGPT and Google results (Chen et al., 2025) means most content that performs well in one channel does not automatically perform well in the other.

6.5 Policy

Three issues deserve regulatory attention. First, Google bundling AI Overviews with its dominant search platform, and rapidly integrating advertising into those overviews, raises antitrust questions. Second, LLMs training on and synthesizing publisher content without compensation is not sustainable for ad-dependent publishers; IAB Europe's 2025 standards offer a starting framework but lack enforcement teeth. Third, if GEO requires significant investment, smaller publishers may be systematically disadvantaged in AI-mediated search, potentially reducing information diversity.

6.6 Limitations

I should be candid about what this paper cannot do. The quantitative data comes from industry reports with different methodologies and potential biases; there is no unified measurement standard for AI referral traffic. The contradictory

zero-click evidence (Section 5.2) remains genuinely unresolved. The qualitative analysis relies on secondary documents rather than primary interviews, so I may be missing important nuances in how publishers actually make decisions. The geographic scope is limited to the US. SEDM's predictions are untested hypotheses. And there is a real possibility that vendor-published data systematically overstates AI referral traffic, which would make SEDM's Phase 2 assessment premature.

7. FUTURE RESEARCH

The most important next step is controlled experimental work that extends GEO-bench to real-world settings with randomized content modifications and longitudinal tracking. The field also needs primary interview data from publisher decision-makers, cross-market comparisons (particularly given EU AI Act implications), and causal econometric modeling of publisher sustainability scenarios. Standardized GEO metrics, analogous to Domain Authority for traditional SEO, would make cross-study comparison possible. And the zero-click paradox from Section 5.2 specifically calls for a controlled before-and-after design with carefully matched keyword sets.

8. CONCLUSION

The LLM disruption of search is real, significant, and far more complicated than either camp in the debate acknowledges. The data is sometimes contradictory: 83% zero-click rates for AIO queries coexist with slightly lower zero-click rates when you track the same keywords over time. A 34.5% CTR decline at the top of search results coexists with 394% growth in AI Overview advertising. An aggregate 2.5% traffic decline conceals 26–38% losses for news publishers alongside barely perceptible changes for e-commerce.

SEDM offers a way to make sense of these tensions by integrating three theoretical traditions that individually capture only part of the picture. The model's five predictions give it the falsifiability that distinguishes a testable framework from a post-hoc narrative. Whether those predictions hold will determine whether SEDM is a useful contribution or an artifact of a particular moment.

What is clear, regardless of theoretical framing, is that the roughly \$2 billion in annual publisher advertising losses is not evenly distributed, is likely growing, and demands both practical adaptation and policy attention. GEO, backed by Aggarwal et al.'s controlled evidence of 30–41% visibility improvements, offers the most promising path for publishers. But it is a path being built while the ground shifts.

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the application of Christensen's framework to the search ecosystem. The DataForSEO Trends API data was collected directly by the author between April 2025 and March 2026. The author thanks anonymous colleagues in the SEO industry who provided feedback on earlier drafts.

DECLARATION OF INTEREST

The author is employed by AT&T in a technical product management role focused on AI and search infrastructure. This research was conducted independently and does not represent AT&T's views, policies, or positions. The author has no financial relationship with Semrush, Chartbeat, DataForSEO, Previsible, or any other data provider cited in this study. DataForSEO API access was obtained through a standard commercial subscription. No external funding was received for this research.

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APPENDIX A: QUALITATIVE DATA SOURCES

The following 23 publisher case studies and strategy documents were analyzed using Braun and Clarke’s (2006) six-phase thematic analysis framework. Documents are listed in order of analysis.

1. Business Insider (2025). “Our AI traffic strategy: 2025 and beyond.” Internal strategy memo, published excerpt.
2. The New York Times (2025). “Navigating AI search: Editorial and SEO alignment.” Public remarks, NYT Open conference.
3. DMG Media (2025). “AI Overviews impact assessment: Q3 2025.” Industry presentation, Digiday Publishing Summit.
4. Dotdash Meredith (2025). “Content optimization for generative search.” Investor day presentation excerpt.
5. Healthline Media (2025). “E-E-A-T and AI citations: Medical content adaptation.” Published case study, Search Engine Journal.
6. Forbes (2025). “Forbes Advisor AI visibility strategy.” Conference presentation, Pubtelligence.
7. Condé Nast (2025). “Multi-platform search strategy update.” Internal memo, reported by Press Gazette.
8. BankRate / Red Ventures (2025). “Financial content GEO implementation: Results and roadmap.” Webinar presentation.
9. WebMD (2025). “Structured data and AI citation tracking.” Published case study, Search Engine Land.
10. Wirecutter / NYT (2025). “Product review optimization for AI referral.” Internal strategy document, excerpted in NiemanLab.
11. Vox Media (2025). “AI traffic diversification: Beyond Google.” Industry talk, SXSW Interactive.
12. The Atlantic (2025). “Subscription-first in the age of AI search.” Public strategy statement.
13. Ziff Davis (2025). “Tech content adaptation: Schema.org and LLM accessibility.” SEO conference presentation.
14. Future plc (2025). “AI-driven traffic: Measurement and response.” Investor relations Q&A.
15. Wired / Condé Nast (2025). “Long-form journalism and AI citation patterns.” Internal analysis, excerpted in Columbia Journalism Review.
16. NerdWallet (2025). “GEO for financial comparison content.” Published case study, Moz Blog.
17. RecipeTin Eats (2025). “Independent publisher AI strategy: Schema, structure, and survival.” Blog post and industry interview.